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Job Quality and Labour Market Transitions: Evidence from Mexican Informal and Formal Workers

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ABSTRACT *We document job characteristics for young, male, urban workers in Mexico, a country with high informal employment and increasing education levels. The informal sector is composed of two distinct parts: salaried informal employment and self-employment. On almost every measure, including wages, informal salaried jobs are of lower quality than formal salaried or self-employed ones. We characterize short-term job type transitions among these workers and show that education plays a key role when transitioning into the formal sector, whereas age is more strongly associated with transitions into self-employment. Persistence in and transitions into formal jobs are more likely for more educated workers. These workers also benefit from higher wage gains when this transition is from informal salaried jobs. On average, wages are higher for workers transitioning into self-employment, but less-educated workers benefit more. For these workers, self-employment can represent an outlet for entrepreneurial talent for some, but like informal salaried work, for others, it can be the sector of last resort.*

KEYWORDS: Informal work; job quality; transitions; Mexico

JEL CLASSIFICATION CODES: J21; J3; J46; O17

1. Introduction

Informal work in Mexico accounts for the majority of employment among all workers (~60 per cent), and a non-trivial (42 per cent in 2017) amount among young, male, urban workers. In recent years there has been a steady increase in educational attainment among this group, yet the proportion of workers in informal salaried jobs has remained steady. These two patterns of higher educational attainment and stagnant proportion of informal, salaried workers raise questions about the role of education and the

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ability of formal labour markets to absorb the increase in better-educated workers (Levy & López-Calva, 2020; OECD, 2019).

To understand the links between education and informal work we examine if transitions out of informality and the benefits of doing so vary by education. To do this we classify three types of jobs: formal salaried, informal salaried, and self-employed.[1] Similar to other papers (Bobba, Flabbi, Levy, & Tejada, 2021; Cano-Urbina, 2015) we show that the informal sector in Mexico is split into two distinct parts: salaried informal employment and self-employment. This duality applies to both more and less-educated workers, highlighting that education does not eliminate the existence of either type of informal employment. Unlike other papers, however, we compare both types of informal jobs to each other and to formal ones.[2] These comparisons show that on almost every measure informal salaried jobs are of lower quality than formal salaried or self-employed ones. Meanwhile, the story of self-employment is more complex. Although average wages are on par with those for salaried formal work, the variance is also greater, manifesting itself in a greater percentage of low- and high-wage workers. This confirms earlier research and debate in the literature that finds that self-employment in Mexico is divided between jobs of opportunity for some and jobs of last resort for others (Cunningham & Maloney, 2001; Fields, 1990; Perry et al., 2007). There is less evidence, however, of this bifurcation within informal salaried work.

We focus on young, urban, male workers (20–40 years old), who are less well-established in the labour market and more likely to transition between job types.[3] Young workers also capture the increase in educational attainment more than older workers, which may better represent how trajectories across jobs vary by education level. Our analysis of transitions among these workers shows clear differences by education level. Better educated workers are more likely to start in formal jobs, more likely to persist in them, and more likely to move into them once in the labour market. They also experience greater wage benefits from transitions into formality and experience larger losses when exiting it. This differs from earlier work by Pagés and Stampini (2009) who find no differences by education level in the degree of mobility from informal salaried work to formal work. They take this as an indication that labour market segmentation applies to all skill levels. Our results, however, suggest this segmentation is less visible for well-educated workers. For these workers transitions into formal salaried work happen early in their careers and are more likely to take place from informal salaried work than from self-employment. This suggests informal salaried work is a stepping stone into formal salaried work for some workers, but more so for better-educated ones.

In explaining the role of informal salaried work as a stepping stone, we find evidence favouring the hypothesis that restrictive labour laws make employers reluctant to hire workers with formal contracts without an initial probationary period. In a sub-sample of individuals with information on job tenure, ∼40 per cent of informal to formal transitions happen within the same employer, and the rate is higher for well-educated workers than for less-educated ones. We also explore the hypothesis that formal employment is a stepping stone to self-employment, such as being an entrepreneur, by providing individuals with the skills and capital to start their own firms. There is mixed evidence of this among the young cohort. On the one hand, transitions into self-employment increase as workers age, and average wages rise following a shift into this type of employment. This supports a story of self-employment as a sector of opportunity. On the other hand, better-educated workers are less likely to shift into self-employment and the bulk of these transitions occur from informal salaried work rather than formal salaried work. In addition, the wage gains from shifting from formal salaried work into self-employment are smaller for better-educated workers. Thus the group with the greatest level of skill and capital is the least likely to move into and benefit from self-employment. This suggests self-employment is not a sector of opportunity for all, and that transitions into it are not always voluntary.

Our study contributes to research and policy discussions on the informal labour markets in Mexico specifically (Cano-Urbina, 2015, 2016; Duval Hernández, 2020; Levy, 2008; Maloney, 1999; 2004) and developing countries more broadly (Kanbur, 2017). In the case of Mexico, our work follows that of Cano-Urbina (2015), who studies the effect of the informal sector on the career prospects of less-educated workers. He finds that in addition to the skill accumulation that may occur, the informal sector plays an important role in screening young, new, less-educated workers. We also find evidence of this, but more for well-educated workers. This likely is because screening and knowledge acquisition has more salience in explaining informal to formal job transitions for this group. Cano-Urbina (2016) also finds that for young less-educated workers the heightened competition of the informal labour market results in higher wage growth than for the same group of workers in the formal sector. Our work suggests this may not be the case for all informal jobs, and that wage trajectories might look quite different for informal salaried and self-employed workers.

In the case of the broader discussion of informality, our contribution is multi-fold. First, the comparison of job types reveals a more complicated story of informal work, as informal salaried work and self-employment emerge as very distinct. Improving informal jobs is an important policy goal, but the appropriate recommendations vary by type of informal job. For example, business training and microcredit programmes would help the self-employed but would have a more limited and indirect impact on informal salaried workers. Similarly, reforms that improve firms' ability to hire temporary workers or fire permanent ones may help reduce informal salaried work, but likely will have a smaller impact on the self-employed. Second, we clarify the role of education in initial job placement and in job-to-job transitions for younger workers. In countries where informality is high, like many Latin American countries (Perry et al., 2007), even with rising education levels more educated workers may find that their initial job is in the informal sector. We show that while more educated workers are much more likely to start in the formal sector, a non-trivial percentage starts in the informal sector as well. Studying these initial job placements is valuable, as the long-term consequences of labour market entry are well-documented for developed countries (Kahn, 2010; Oreopoulos, von Wachter, & Heisz, 2012), but less so for developing countries.

In investigating persistence, we find that more educated workers who start in the informal sector are more likely to transition into formal employment. Less-educated workers are less likely to exit informality and are more likely to shift from self-employment to informal salaried work, and vice versa. They also are more likely to exit formal salaried work into lower quality, informal salaried work. This is contrary to the idea of voluntary participation in the informal sector (Maloney, 2004), and suggests some workers are stuck in the informal sector. This confirms that path dependence on informality may matter in Mexico, as pointed out by Levy (2008). The inability of workers to leave the informal sector may have long-term implications for social security contributions and the provision of benefits to formal and informal workers.

The rest of the paper is structured as follows: in Sections 2 and 3 we describe the data used in this study, including a discussion of the job type characteristics of young Mexican workers; in Section 4 we use multinomial logit regressions and transition matrices to describe workers' transitions in and out of informal work, how those transitions vary by education and across and within employers, and changes in wages that result from these shifts; and in Section 5 we summarize our main findings and provide reflections on potential policy implications.

2. Data

For our analysis, we use the *Encuesta de Ocupación y Empleo* (ENOE), a rotating labour force survey conducted by the National Institute for Statistics and Geography (INEGI).[4] The ENOE started in 2005 and follows individuals in households for up to five quarters. We use the quarterly surveys from the first quarter of 2005 to the fourth quarter of 2017 and focus on young workers, defined as individuals who are between the ages of 20 and 40 when they enter the sample and are employed. We also limit the sample to men in urban areas, to abstract from differences in labour force participation and job types in primary industries like agriculture.[5] While we exploit the panel nature of these data to examine short-term transitions from school to work and across employment types, for the summary statistics we construct a cross-section such that each individual only appears once. To do this we use individuals' responses as of the last interview.

We define a job as formal if an individual is registered with the social security administration (IMSS) or equivalent agency (the military, Pemex, or state social security agency), which is required by Mexican labour regulations for workers with a contract.[6] This definition is used by the ENOE to classify jobs as formal (ENOE, 2014), and has been used by other researchers as well (Cano-Urbina, 2016, Bobba et al., 2021). The information on registration status allows us to categorize three types of jobs: formal salaried jobs, informal salaried jobs, and self-employed.[7] Of these, we categorize formal salaried work as formal and informal salaried and self-employed as informal, since 99 per cent of self-employed workers in our sample are not registered with any type of social security programme. This means the informal sector is made up of two job types while the formal sector is made up of one.

3. Job type characteristics

In this section, we document the characteristics and trends of formal and informal jobs. We describe formal salaried jobs in relation to both types of informal jobs: salaried and self-employment. Figure 1(A) shows the percentage of male, urban workers aged 20–40 who fall into each job type. While formal salaried work has the highest proportion of workers as a single category, informal salaried work and self-employment combined account for a non-trivial proportion of employment. For example, by the end of 2017 informal salaried work and self-employment account for 42 per cent of all jobs for young workers. Among these, informal salaried work has the highest proportion, accounting for 25 per cent of all young, male, urban workers in 2017, compared with 17 per cent for self-employment. This shows that in Mexico considering only self-employment ignores the largest portion of the informal sector—informal salaried work. The graph also shows that formal salaried work has increased by about three percentage points from 2005 to 2017, while informal work has declined by the same amount. The decline in informal work is entirely driven by self-employment, as informal salaried work is approximately the same in 2005 and 2017.

Figure 1(B) shows the trajectory of average years of education for each job type. As expected, average education levels are significantly higher for formal salaried jobs than for either type of informal work. For example, at the end of 2017 average education was 12.4 years for formal salaried jobs, 11.4 years for self-employment, and 10.4 years for informal salaried jobs. This means the average formal worker has slightly more than an upper secondary education, while the average informal worker (both job types) has some upper secondary education (more than nine years). The figure also shows a steady increase in education levels for all types of employment, a result of rising educational attainment in Mexico (OECD, 2019). For formal salaried workers, average education levels rose from slightly less than an upper secondary education (11.1 years) to slightly more (12.4), while for informal work (both types) the increase added more years of upper secondary education.

Table 1 provides a summary of job characteristics by job type, including real, hourly wages (in pesos); working more than 20 and <45 h per week (in other words, is not under- or over-employed); has a written contract; receives a bonus; receives paid vacation days; and is entitled to profit-sharing.[8] The table also includes two job quality indices. The first ranges from 0 to 1 where a higher value indicates better job characteristics.[9] The second index uses principal component analysis and includes the same variables as in the previous index. Both indices indicate that formal jobs are better than salaried informal. The first principal component has a variance

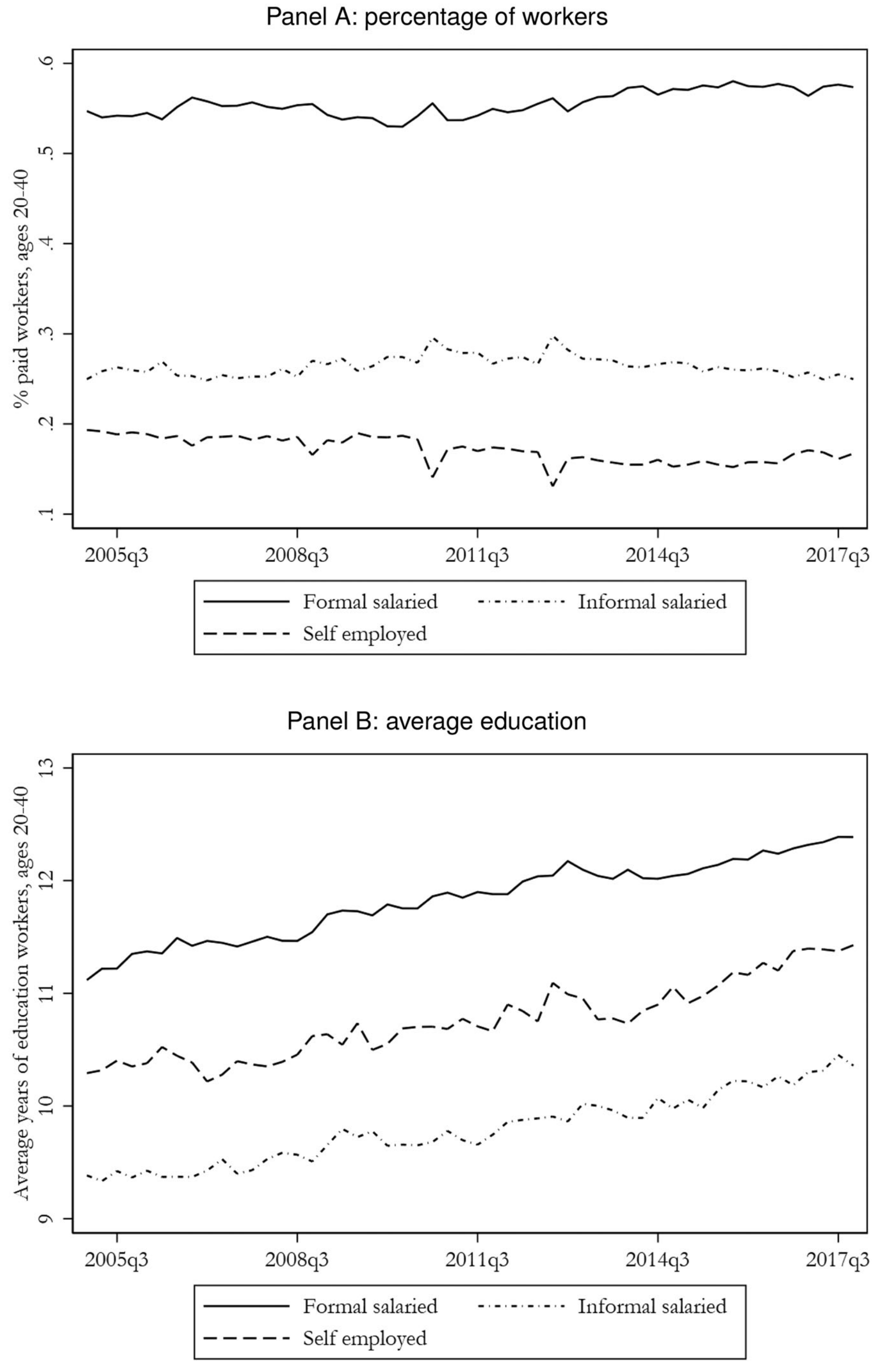

Figure 1. Composition of job types. (A) Percentage of workers. (B) Average education.
Note. Male, urban workers are aged 20–40.
Source: authors' compilation based on ENOE Q12005–Q42017.

of 2.72, explaining 45 per cent of the total variance; it is also highly correlated with the aggregated job quality index (correlation coefficient of 0.96).

To see if job characteristics differ by education level, we split the sample into two groups: less-educated workers with primary education or less (six years or less) or lower secondary education (nine years); and more educated workers with an upper secondary education (12 years)

Table 1. Job characteristics

	Primary and lower secondary			Upper secondary and college		
	(1) Salaried formal	(2) Salaried informal	(3) Self employed	(4) Salaried formal	(5) Salaried informal	(6) Self employed
Job characteristics						
Hourly wage (pesos)	19.63 (11.64)	17.83 (14.64)	25.93 (25.98)	34.02 (29.97)	25.62 (26.20)	40.58 (41.46)
Weekly hours	49.49 (13.79)	49.66 (16.72)	46.37 (19.66)	45.97 (13.87)	43.93 (17.71)	45.08 (19.30)
Weekly hours >20 and <45	0.29 (0.45)	0.26 (0.44)	0.30 (0.46)	0.43 (0.49)	0.37 (0.48)	0.34 (0.47)
Has written contract	0.87 (0.34)	0.08 (0.26)	. (.)	0.94 (0.24)	0.25 (0.43)	. (.)
Bonus	0.92 (0.27)	0.19 (0.39)	. (.)	0.95 (0.21)	0.26 (0.44)	. (.)
Paid vacation days	0.86 (0.35)	0.09 (0.28)	. (.)	0.92 (0.28)	0.18 (0.38)	. (.)
Profit sharing	0.39 (0.49)	0.01 (0.11)	. (.)	0.36 (0.48)	0.03 (0.16)	. (.)
Job quality index [0–1]	0.72 (0.19)	0.26 (0.15)	. (.)	0.76 (0.16)	0.33 (0.21)	. (.)
Job quality PCA index	0.86 (1.06)	−2.08 (0.89)	. (.)	1.03 (0.82)	−1.69 (1.22)	. (.)
Observations	100,378	68,283	37,772	130,647	37,684	32,686

Notes: Population-weighted means reported. Standard deviations in parentheses. Cross-section of male, urban workers aged 20–40.
Source: authors' compilation based on ENOE 2005–2017.

or college and beyond (16 years or greater). An open question is whether informal jobs look different for this group relative to less-educated workers.[10]

There are three key takeaways from this table. First, education is strongly correlated with formal jobs: the proportion of more educated workers in formal jobs is 65 *vs.* 49 per cent for less educated. Second, informal salaried work has lower average quality than either formal salaried work or self-employment on every measure other than hours. Starting with wages, for well-educated workers, average wages for informal salaried are 30 per cent lower than formal salaried, while for less-educated workers they are 14 per cent lower. We see even larger gaps in non-wage job characteristics. Informal salaried workers are 2.5–4.5 times less likely to receive a bonus, 4.5–11 times less likely to receive paid vacation, and 14–37 times less likely to receive profit sharing. The job quality index for informal salaried work is 154 per cent lower than for formal salaried among the less educated and 114 per cent lower among the better educated. This last statistic shows that education is not enough to make informal salaried jobs better than formal ones. Third, when we look at self-employment there is less evidence of lower quality since average self-employed wages are slightly higher than formal salaried ones for both education groups. The basic wage comparisons, however, suggest some self-employed jobs may be on par with salaried formal ones.

We also examine the distribution of hourly wages by job type. We divide all wages into quartiles and calculate the percentage of each job type that falls into each quartile. These results are presented in Figure 2(A) and show distinct patterns: salaried formal work and salaried informal work have opposing patterns. Salaried formal work shows the highest percentage of workers in the highest wage quartile and the smallest percentage in the lowest wage quartile with steady increases in proportions as wages rise. Meanwhile, salaried informal work has the highest percentage in the lowest wage quartile and the lowest percentage in the highest wage quartile, with a sharp decline in proportions as wages rise. This is consistent with the previous

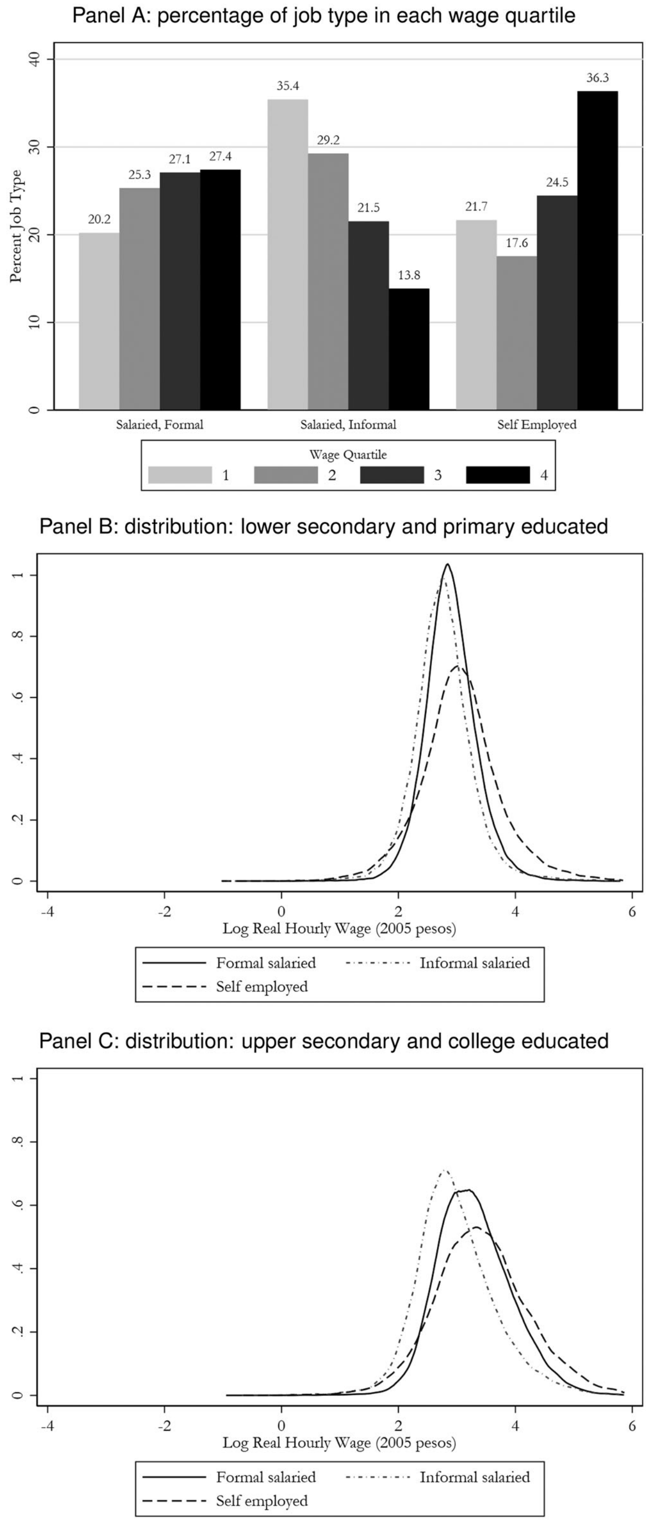

Figure 2. Wages. (A) Percentage of job type in each wage quartile. (B) Distribution: lower secondary and primary educated. (C) Distribution: upper secondary and college educated.

Notes: Cross-section of male, urban workers age 20–40. Wages in Q12005 pesos. Top 0.1 per cent trimmed.

Source: authors' compilation based on ENOE Q12005–Q42017.

characterization of informal salaried jobs being lower quality. The same cannot be said, however, of self-employment. This exhibits a different pattern entirely, with a non-monotonic distribution of workers as wages increase. The highest proportion of self-employed workers is in the highest wage quartile, followed by the lowest quartile, aligning with a story of duality in which self-employment represents an outlet for entrepreneurial talent for some and the sector of last resort for others.

The wage results paint informal salaried work as a sector where wages are lowest but relative to self-employment it may offer more job security. As shown in Figures 2(B) and 2(C), the three earnings distributions of self-employed, formal, and informal salaried workers substantially overlap. However, the distribution of wages for informal salaried work is centered to the left of the distribution of formal salaried and self-employed workers. This implies the median earnings for self-employed and formal workers are higher than for informal employees. This pattern is found in both well-educated and less-educated workers. Also for all workers, wage dispersion is highest for the self-employed, indicating that for some, this is a riskier option than those offered by salaried jobs.

In general, the findings in this section suggest informal salaried work and self-employment may play different roles in the working lives of individuals. For example, for some workers, informal salaried work and self-employment are jobs of last resort, while for others self-employment, specifically, may be an optimal choice, providing higher wages and job flexibility. Yet for others, informal salaried work may provide more stability or be a necessary first stage before transitioning to formal salaried work. This would occur if restrictive labour legislation makes employers reluctant to offer formal contracts to new workers; or if employers instead start workers on informal contracts and move them into formal ones after a probationary screening period. We seek to illuminate each of these possibilities in the next section.

4. Labour market transitions

We turn to labour market transitions to examine if and when young workers shift across job types to better understand the role that informal salaried work and self-employment play. We are particularly interested to see if transitions out of informality and the benefits of doing so vary by education and if either has changed over time.

The ENOE is a short panel in which individuals, at most, appear in five quarters. While the ENOE is not long enough to see the evolution of individuals' working lives, we can see quarterly transitions for those who change jobs within this time frame.[11] Exploiting the large cross-sectional dimension of the ENOE to infer transition rates, we graph the percentage of young workers by education level that fall into each job type by age. These are presented on the left-hand side of Figure 3 and show substantial differences by education level. As shown in panel A, ~46 per cent of less educated, male, urban workers of age 20 are in informal and formal salaried work, while only 7 per cent are self-employed. As workers age, both formal and informal salaried work decline, while self-employment rises. This rise is large (twenty percentage points), and by the age of 40 self-employment has more workers than informal salaried work. This suggests that for less-educated workers the transitions into formality are low and that more change happens by shifting across informal sector employment.

As shown in Figure 3(B), we see a similar, but slightly different story for more educated workers. At age 20 salaried formal work also is dominant, making up 55 per cent of all male, urban workers. Informal salaried work makes up 38 per cent of workers and self-employment only 6 per cent. Similar to less well-educated workers we see that informal salaried work declines in age, while self-employment rises. What is different for this group, however, is that formal salaried work rises as well. The shift into formality change happens early on and quickly, and by age 28, 69 percent of more educated, urban male workers have formal, salaried jobs. These comparisons show that the likelihood of becoming a formal worker conditional on

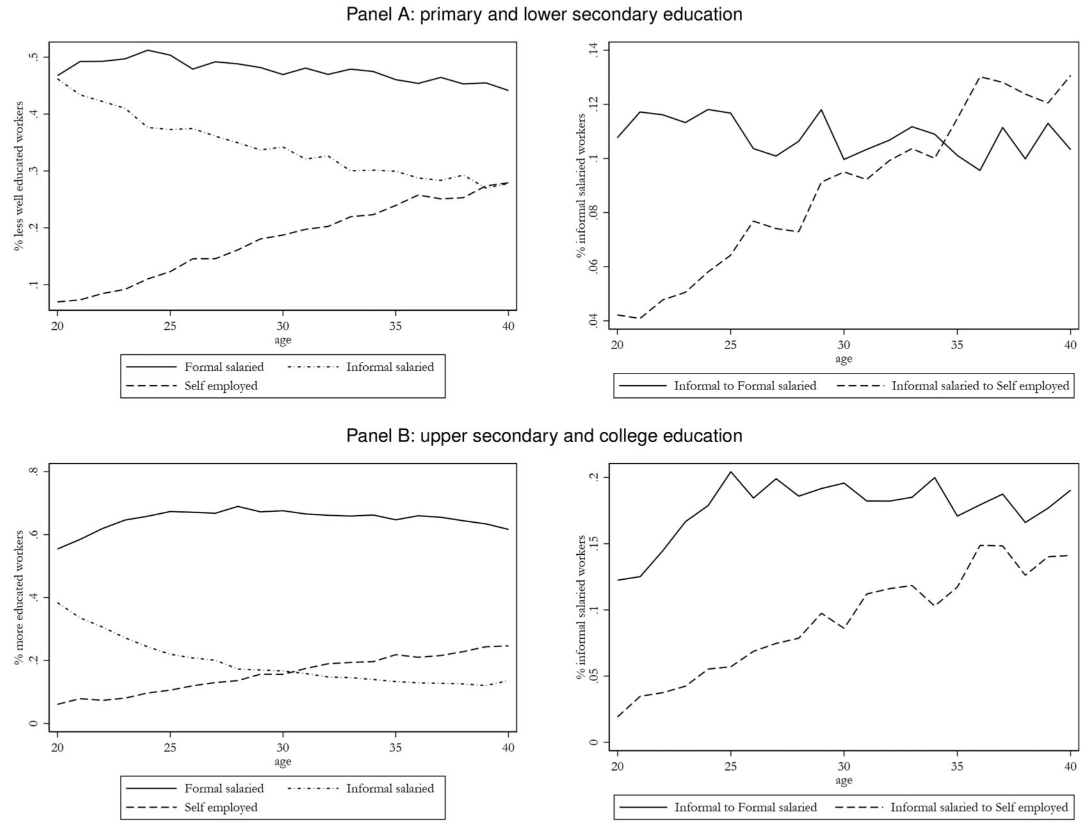

Figure 3. Job type, transitions, and age. (A) Primary and lower secondary education. (B) Upper secondary and college education.

Notes: Male, urban workers are aged 20–40. Transition rates are defined as the percentage of informal salaried workers in one quarter who transition to salaried formal work or self-employment in the next quarter. These quarterly rates are averaged across age groups.

Source: authors' compilation based on ENOE Q12005–Q42017.

age is stronger among this better-educated group of workers. For less-educated workers the likelihood of transitioning to a formal job decline faster as they age.

The left-hand side figures of Figure 3 provide a sense of the transition dynamics using the cross-section. To ensure we are not confounding the cohort effect with the time effect, we turn to the panel dimension of the ENOE to calculate and graph quarterly rates of informal salaried worker transitions into formal salaried jobs or self-employment. As shown in the graphs on the right-hand side of Figure 3, for both education levels the transition from informal salaried work to self-employment steadily rises over time and follows a similar trajectory. This is consistent with salaried work, both formal and informal, providing workers with skills that they can apply to self-employment. A key difference across education levels, however, is that less-educated workers older than 34 have a higher rate of transition into self-employment than formal salaried work, whereas for better-educated workers the transition rate into formality is always higher than the one into self-employment. This suggests informal salaried work plays a different role in the working lives of less-educated workers relative to better-educated ones, and we explore this in more detail below.

4.1. Transition from school to work

We look at the determinants of labour market outcomes for individuals who transition out of school and into work during the time they are in the ENOE. To do this we track

Table 2. School-to-work transition: multinomial logit

	Relative to salaried informal (base)			
	(1) Salaried Formal	(2) Self Employed	(3) Salaried Formal	(4) Self Employed
Age	0.0012 (0.0019)	0.0066*** (0.0011)	0.0014 (0.0019)	0.0065*** (0.0011)
Married	0.0263 (0.0245)	0.0022 (0.0160)	0.0252 (0.0245)	0.0024 (0.0160)
Head of HH	0.0410 (0.0277)	0.0219 (0.0164)	0.0409 (0.0276)	0.0221 (0.0163)
HH size	0.0072 (0.0046)	−0.0023 (0.0031)	0.0069 (0.0046)	−0.0022 (0.0031)
Lower secondary	0.1118*** (0.0360)	−0.0532** (0.0242)	0.0004 (0.0672)	−0.0303 (0.0447)
Upper secondary	0.1474*** (0.0333)	−0.0258 (0.0204)	0.0421 (0.0620)	−0.0006 (0.0381)
College	0.2170*** (0.0362)	−0.0208 (0.0212)	0.1762** (0.0692)	0.0109 (0.0404)
Linear Time Trend	0.0000 (0.0076)	0.0011 (0.0053)	−0.0043 (0.0077)	0.0019 (0.0054)
Lower Sec. × Time Trend			0.0046** (0.0022)	−0.0010 (0.0015)
Upper Sec. × Time Trend			0.0043** (0.0020)	−0.0011 (0.0012)
College × Time Trend			0.0021 (0.0022)	−0.0013 (0.0013)
Observations	10,541	10,541	10,541	10,541
Baseline Predicted Probability	0.64	0.12	0.64	0.12

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Standard errors in parentheses. Population-weighted, average marginal effects from multinomial estimations are reported. The base group is salaried informal workers. The base educational category is primary education or below. Other controls include industry, state, year, and quarter fixed effects. Panel of male, urban workers are aged 20–40.
Source: authors' compilation based on ENOE 2005–2017.

individuals who are matriculated in school when they enter the panel but are working in their last interview, regardless of how many quarters they are in the sample.[12] This ensures consistency across individuals and no double counting. While we do not know if the recorded jobs are the individuals' first ones, these individuals likely are at the earliest stages of their working lives. We thus argue this group comes close to telling us about the determinants of initial job placement.

We estimate the probability of salaried formal employment and self-employment relative to salaried informal employment using a multinomial logit model. The model assumes no nesting, thereby allowing us to consider all job types equally. We think this is a more realistic scenario, where individuals consider all three job types simultaneously, rather than one in which workers first choose informality and then decide to sort into informal salaried or self-employment.

Within the multinomial logit model, the probability that person i falls into job type j can be written as:

$$pr_{ij} = \left(\frac{\exp\left(X_i'\beta_j\right)}{\sum_{k=1}^{m}\exp\left(X_i\beta_l\right)}\right) \tag{1}$$

For observable characteristics (X) we include age, whether or not a person has a spouse or partner, whether the person is the head of the household, the household size, education, and industry, state, year, and quarter fixed effects. We also include a linear time trend and this trend interacted with different education levels. This highlights a benefit of using 13 years of data from the ENOE, as it allows us to see if sorting across job types has changed over time. Our base group is informal salaried employment and the coefficients in the table are average marginal effects. Each coefficient, therefore, represents how a one-unit increase in the variable changes the probability that an individual is in formal salaried employment or self-employment relative to informal salaried employment.

We present average marginal effects and their standard errors in Table 2. The predicted probability of formal salaried work is 0.64 (column one), in contrast to 0.12 for self-

Table 3. Job type transitions

	(1) Primary	(2) Lower secondary	(3) Upper secondary	(4) College
Were salaried formal				
Stays Sal. formal	0.81	0.89	0.90	0.91
Becomes Sal. informal	0.15	0.09	0.08	0.06
Becomes self employed	0.03	0.02	0.02	0.03
Were salaried informal				
Becomes Sal. formal	0.14	0.18	0.24	0.36
Stays Sal. informal	0.73	0.69	0.63	0.50
Becomes self employed	0.13	0.12	0.12	0.13
Were self employed				
Becomes Sal. formal	0.05	0.07	0.09	0.13
Becomes Sal. informal	0.25	0.22	0.17	0.11
Stays self employed	0.70	0.71	0.74	0.76
Obs. begin Sal. formal	66,094	203,442	181,089	163,778
Obs. begin Sal. informal	71,710	94,848	58,385	30,079
Obs. begin self employed	39,828	56,289	42,676	38,056

Notes: Standard errors in parentheses. Population-weighted averages are shown. Transitions are calculated from one quarter to the next, conditional on remaining employed. Panel of male, urban workers are aged 20–40.
Source: authors' compilation based on ENOE 2005–2017.

employment (column two), and 0.24 for informal salaried work (by default the remaining percentage). The higher probability of formal salaried work is consistent with our sample of workers as depicted in Figure 3. Meanwhile, the low predicted probabilities of self-employment stem from the fact that fewer individuals, regardless of education, start their working lives in this job type.

Column one of the table shows that relative to informal salaried there is a strong education gradient for transitions into salaried formal employment. The likelihood of this transition is highest for college-educated workers and it steadily declines with the education level. In contrast, the coefficient on education becomes negative and sometimes insignificant in the transitions from salaried informal to self-employment, and what matters most in this transition is age, where older workers are more likely to become self-employed. This may suggest that skills and capital acquired through salaried work may be more important than education in determining self-employment.

The general time trend is insignificant in columns one and two, indicating that the school-to-work transitions into a specific job type do not appear to change over time. When interacting with school level, the time trends become positive for lower and upper secondary (column three), which suggests there has been an increase in the number of workers in the middle of the education distribution who start with formal jobs, but not an increase in the number of college-educated ones who do. However, these effects are small.

4.2. Transitions within the labour market

We estimate quarterly transitions across job types once individuals are in the labour market. Summary statistics on changes are shown in Table 3 and show clear differences in education in the persistence of and transitions out of informality. We note that the number of individuals who start in each job type by education level is provided at the bottom of the table.

Starting with formal salaried work, the table shows that transitions into this job type and persistence in it increase educational attainment. For example, as shown in the fourth row, 36 per cent of college-educated workers in salaried informal jobs transition to salaried formal ones, compared with only 14 per cent of primary educated workers. The same values of switching from self-employment to salaried formal work are 13 *vs.* 5 per cent (seventh row). This corresponds to around 40 per cent higher transitions into formality for more educated workers relative to the least educated ones. Meanwhile, 91 per cent of college-educated, salaried formal workers stay in that job type across quarters, while only 81 per cent of primary educated workers do. These comparisons indicate that informal work does provide a stepping stone into formal work for some, but more so for well-educated workers.

Second, flows out of the formal sector and into the informal exist, but are driven by informal salaried work instead of self-employment. For example, among formally employed less-educated workers, between 9 and 15 per cent transfer to informal salaried work, and only 2–3 per cent transfer to self-employment. Among well-educated workers, these values are 6–8 and 2–3 per cent, respectively. This contradicts a story of voluntary exit from formality driven by workers who use skills and capital gained in the formal sector to open their own firms. This may be because younger workers still face barriers to self-employment and the transition rates are higher among older workers. Why so many workers fall out of formal salaried work and into informal salaried work is unclear, and warrants further investigation.

Third, the rates of persistence are the highest for formal salaried work, with 90–91 per cent of well-educated workers and 81–89 per cent of less-educated workers staying in formal salaried jobs across quarters. This is unsurprising as better job characteristics and restrictive labour legislation make job separation less likely. Rates of persistence in informal salaried jobs are lower for well-educated workers (50–63 per cent) than less educated ones (69–73 per cent), providing further support for the idea that they serve as more of a stepping stone to formality for the former over the latter.

Fourth, the rates of persistence in and the type of transition out of self-employment vary across education groups. Persistence rises gradually with education, with 76 per cent of college-educated self-employed workers staying across quarters compared with 70 per cent of primary educated ones. Of those who transition, for primary educated workers the majority shift to informal salaried work, while for college-educated workers the majority transition to formal salaried work. In other words, many less well-educated workers stay in the informal sector, while better-educated workers are more likely to exit it. It is unclear if these are voluntary transitions driven by better informal job opportunities or involuntary ones driven by skill, capital, or labour constraints.

Table 4 further explores transitions by estimating movements *out of* informal salaried work or self-employment relative to remaining in either type of work. We again estimate a multinomial logit model, using staying in informal salaried work (columns one and two) or self-employment (columns three and four) as the base. The controls are the same as in the previous model and the coefficients reported are average marginal effects.

The results reported in Table 4 confirm that the likelihood of transitioning out of either type of informal work and into formal salaried work increases significantly in education. For example, as shown in column one, lower secondary educated workers are only 3.99 per cent more likely than primary educated workers to move from informal salaried work to formal salaried work while college-educated workers are 23.3 per cent more likely to make the same transition. Meanwhile, as shown in column three lower secondary educated workers are 5 per cent more likely than primary educated ones to transition from self-employment to salaried formal work, while college-educated workers are 8.9 per cent more likely to do so. The comparison across both types of informality shows that transition rates for well-educated workers (upper secondary and college) are two to three times higher for informal salaried work than self-employment. Meanwhile, the coefficients on education are negative for transitions out of self-employment and into informal salaried work. This further suggests that informal salaried work is more of a sector of last resort than self-employment, particularly for workers with more education.

The table also shows that older workers are more likely to stay in their sector and less likely to transition into the formal sector, yet we see that as people age they are more likely to transition

Table 4. Transitions from informal job types: multinomial logit

	Relative staying informal Sal.		Relative staying self employed	
	(1) Salaried formal	(2) Self employed	(3) Salaried formal	(4) Salaried informal
Age	−0.0033*** (0.0003)	0.0076*** (0.0002)	−0.0028*** (0.0002)	−0.0084*** (0.0003)
Married	0.0300*** (0.0041)	0.0027 (0.0037)	0.0063 (0.0039)	−0.0109** (0.0049)
Head of HH	−0.0338*** (0.0044)	0.0123*** (0.0038)	−0.0177*** (0.0036)	−0.0336*** (0.0049)
HH size	−0.0021*** (0.0007)	0.0014* (0.0007)	−0.0004 (0.0008)	−0.0011 (0.0010)
Lower secondary	0.0399*** (0.0080)	−0.0040 (0.0069)	0.0508*** (0.0071)	−0.0608*** (0.0088)
Upper secondary	0.1177*** (0.0088)	0.0093 (0.0081)	0.0497*** (0.0075)	−0.0582*** (0.0099)
College	0.2328*** (0.0109)	0.0129 (0.0096)	0.0883*** (0.0073)	−0.1400*** (0.0119)
Linear Time Trend	0.0007 (0.0014)	−0.0011 (0.0012)	0.0006 (0.0012)	0.0014 (0.0016)
Lower Sec. × Time Trend	0.0006** (0.0003)	0.0002 (0.0002)	−0.0006** (0.0002)	0.0008*** (0.0003)
Upper Sec. × Time Trend	−0.0001 (0.0003)	0.0003 (0.0003)	0.0005* (0.0003)	−0.0002 (0.0003)
College × Time Trend	−0.0012*** (0.0004)	0.0003 (0.0003)	0.0002 (0.0002)	0.0006 (0.0004)
Observations	221,499	221,499	159,735	159,735
Baseline Predicted Probability	0.24	0.14	0.10	0.20

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Standard errors in parentheses. Population-weighted, average marginal effects from multinomial estimations are reported. The base group is workers who stay informal salaried or self-employed. The base educational category is primary education or below. Other controls include industry, state, year, and quarter fixed effects. Panel of male, urban Workers are aged 20–40.
Source: authors' compilation based on ENOE 2005–2017.

out of informal salaried and into self-employment. In regards to the time trends, we see only minor changes in transitions across time. However, several are important to note. The coefficient on the college time trend is negative in column one, which means college-educated workers are less likely to transition out of informal salaried work and into formal salaried work over time. The coefficient is small but may capture that with an increasing supply of college-educated workers, more are finding it hard to move into formal work. Meanwhile, there is no significant change in self-employment over time, so this sector is not picking up the slack.

In sum, education is associated with increased transitioning into formal work, particularly from informal salaried work, but not with shifts within the informal sector, while age is a main determinant of transition into self-employment.

4.3. *Transitions across and within employers*

We next explore whether there is any indication that informal salaried jobs are used by employers to screen new employees before offering them permanent contracts. We examine if transitions from informal to formal jobs involve changes across or within *employers*. We can do this for individuals who appear in the long-form questionnaire, applied approximately one in every four quarters, which includes questions about how long a person has been with their current employer.[13] For these individuals we therefore can see if an informal to formal transition happened by switching employers or within the same employer, indicating a screening process. These results appear in Table 5 and do show a higher rate of transitioning within employers for upper secondary and college-educated workers. For example, 45 per cent of college-educated workers transition job types with the same employer, compared with only 35 per cent of lower secondary educated workers. Interestingly, however, the rates are fairly similar, and

Table 5. Employers for transitions to formality

	Educational attainment			
	(1) Primary	(2) Lower secondary	(3) Upper secondary	(4) College
Of those who become formal				
Employer is new	0.58 (0.49)	0.65 (0.48)	0.57 (0.50)	0.55 (0.50)
Employer is the same	0.42 (0.49)	0.35 (0.48)	0.43 (0.50)	0.45 (0.50)
Observations	1,936	3,150	2,341	1,569

Notes: Population-weighted averages are shown. Transitions are calculated from one quarter to the next, conditional on remaining employed. Panel of male, urban workers are aged 20–40.
Source: authors' compilation based on ENOE 2005–2017.

Table 6. Changes in log wages

	Transition					
	(1) Informal Sal. to formal Sal.	(2) Formal Sal. to informal Sal.	(3) Self Emp. to formal Sal.	(4) Formal Sal. to SelfEmp.	(5) Informal Sal. to SelfEmp.	(6) Self Emp to informal Sal.
Education level						
Primary or less	0.045 (0.491)	−0.040 (0.506)	−0.139 (0.673)	0.119 (0.621)	0.195 (0.691)	−0.193 (0.694)
Lower secondary	0.047 (0.533)	−0.044 (0.536)	−0.164 (0.749)	0.201 (0.740)	0.176 (0.718)	−0.200 (0.704)
Upper secondary	0.073 (0.603)	−0.072 (0.589)	−0.213 (0.820)	0.150 (0.789)	0.215 (0.768)	−0.230 (0.797)
College	0.081 (0.650)	−0.093 (0.645)	−0.099 (0.871)	0.009 (0.825)	0.149 (0.876)	−0.117 (0.869)
Observations	35,679	33,583	8,590	8,657	20,937	21,490

Notes: Population-weighted averages are shown, with standard deviations in parentheses. Transitions are calculated from one quarter to the next, conditional on remaining employed. Panel of male, urban workers are aged 20–40.
Source: authors' compilation based on ENOE 2005–2017.

non-trivial (42–45 per cent) for workers at the higher (college or more) and lower end of the education distribution (primary education or less). This suggests that while the story of employer screening exists for all groups, it is greater for those at the higher and lower end of the education distribution.

4.4. Changes in wages

To address the benefits or costs of shifting jobs we explore changes in real wages that result from transitions in job type. Table 6 shows the average change in log wages by each transition in job type and reveals clear differences by education level. For example, as shown in column 1, the improvement in wages that results from moving out of informal salaried work and into formal salaried work is almost twice as high for college-educated workers than for those with primary education or less (8.1 *vs.* 4.5 per cent). Meanwhile, the decline in wages that results from the opposite transition is almost twice as high for college-educated workers as well, showing the symmetry in the benefits and losses by education level.

Interestingly, however, the linearity of wage changes by education disappears when we look at other transitions. For example, as shown in column 4, the average increase in wages from moving from informal salaried work to self-employment is the highest for lower and upper secondary educated workers, but low for college-educated ones. This complicates the story of self-employment as a sector of opportunity for those who acquire skills and capital in the formal sector and use them to start their own businesses. For young workers, the results suggest this story is more relevant for those in the middle of the education distribution than those at the upper end of the distribution.

Finally, as highlighted above, the table shows that transitions to self-employment, on average, lead to improvements in wages, even when the transition is out of formal salaried work. Meanwhile, transitions to informal salaried work lead to average declines in wages, with the largest declines coming from shifting out of self-employment and into informal salaried work. This provides further support that informal salaried work is the least appealing along the wage dimension and the reasons why some individuals transition into this work, or remain there, warrant further investigation.

5. Summary and conclusion

In this paper, we analyze formal and informal work among young, male, urban workers in Mexico, a country with high informal employment and increasing education levels. The combination of higher educational attainment with a relatively stagnant proportion of formal workers indicates that education does not entirely protect against a low-quality job in the informal sector (Levy & López-Calva, 2020). To better understand this duality, we analyze transitions in and out of informal work, how those transitions vary by education and the changes in wages that result from them. Our findings indicate that education is strongly correlated with formal and better-quality jobs. Better educated workers are more likely to start in formal jobs, more likely to persist in them, and more likely to move into them once in the labour market. They also experience greater wage benefits from transitions into formality and larger losses when exiting it. Meanwhile, the link between education and transitions into self-employment is less obvious. Transition rates into self-employment are similar across education levels, while the wage benefits are highest in the middle of the education distribution. This suggests that self-employment does not represent a sector of opportunity for all.

Our analysis contributes to the understanding of developing and middle-income economies that have experienced increases in human capital with different quality jobs available to more workers. This has larger implications for the structure of the labour market and the workers' ability to move between job types. Understanding the role of education in formal and informal labour market transitions helps identify which policies might be needed and for whom. In particular, during a period of increasing education attainment, identifying how to facilitate the transition from school or an informal job to a formal salaried job or higher quality self-employment, can prove beneficial to many workers. Our work can inform policy-makers, and reforms that make it easier for firms to hire formal workers or to operate formally could be considered (Fajnzylber, Maloney, & Montes-Rojas, 2011; Piza, 2018). This should be done while accounting for existing social programs, which themselves may have disincentive effects on formal employment (Bergolo & Cruces, 2021). Training programmes could facilitate transitions into self-employment among informal less-educated workers (McKenzie & Woodruff, 2014). Meanwhile, given the role of education in labour market transitions, other reforms, are likely to have a larger effect on transitions into formality among the more educated workers. Nevertheless, among less-educated workers there still could be a non-negligible effect, helping alleviate concerns of rising education levels in the presence of high informality.

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Notes

1. Following the recent literature such as Duval Hernández (2020), we define informal sector jobs as those without social security benefits.
2. Bobba et al. (2021) and Cano-Urbina (2015) do not include self-employed workers.
3. For example, among well-educated workers, the transition rate from informal salaried to formal salaried work is highest before the age of 28.
4. The data and documentation for the ENOE are publicly available on INEGI's website: www.inegi.gob.mx.
5. For the sample of urban men between the ages of 20 to 40 there are 1,968,927 observations on 545,384 individuals. This means the average number of quarters in which a person appears is 3.7. Approximately 56 per cent of the sample appears in all 5 quarters, while 15 per cent appears in only one. The sample size for our analysis is smaller because we only use those who are employed, which make up ~80 per cent of the total. Finally, we also limit the employed sample to those who receive a salary. This means we exclude unpaid workers, who account for ~2.4 per cent of all young, male, urban workers.
6. Social security registration also means a worker has access to state-funded medical services. The determination of social security registration comes from a question on access to medical services.
7. Salaried jobs are defined as those where an individual says they have a boss, while self-employed jobs either do not have a boss or are employers themselves. The latter is how ENOE defines self-employment.
8. We deflate all earnings to be in quarter 1 of 2005 pesos using CPI data from INEGI. We also remove the top 0.1 per cent of wages to account for outliers.
9. Specifically, following Sehnbruch, González, Apablaza, Méndez, and Arriagada (2020), for each salaried worker, formal or informal, we code a value of 1 or 0 for each of the following six variables: (1) works more than 20 and less than 45 hours per week (in other words, is not under- or over-employed); (2) has a written contract; (3) has *not* looked for a job in the past three months; (4) receives a bonus; (5) receives paid vacation days; and (6) is entitled to profit sharing. We add the responses on all values and divide by six, such that the index value represents the percentage of these six characteristics a worker has.
10. The determination of educational attainment is complicated in the ENOE, as there is not a single variable that captures this. We use questions on the highest education level achieved and years completed at that level, which allows us to distinguish those who likely finished a level from those who dropped out. For example, both lower and upper secondary school require three years each. Someone who lists upper secondary school as the highest education level achieved but was there for one year likely is a dropout and is coded as achieving only a lower secondary education. Meanwhile, someone who completed three years at that level likely completed it. For college graduates, we code individuals as having a college degree if they list college (normal, technical, and university) as the highest education level achieved, are in programmes that require an upper secondary education, and have completed the requirements for the career (*carrera*).
11. This question is the only one that captures changes in student status, as the ENOE does not ask if an individual finished school.
12. This question is the only one that captures changes in student status, as the ENOE does not ask if an individual finished school.
13. Specifically, the extended questionnaire was administered in every quarter in 2005 and in the first two quarters of 2006. After that it was administered only once each year; in quarter 2 in 2007 and 2008 and in quarter 1 in 2009 onward.

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